Higher Diploma in Software Engineering (IT114105)
Coursework (Semester 5 - 2025/2026)
Enterprise Systems Development (ITP4511)

Students are required to upload the software project implementation and the related documents to http://moodle.vtc.edu.hk on or before the submission date.

Date of Source and Report Submission: ***Week 35***
Demonstration Period: ***Week 35***

Students are required to submit their work on the Moodle platform and demonstrate their assignment during the lab session. Late submissions will NOT be accepted, and any late assignment will score a zero.

Scenario

The Community Care Health Consortium (CCHC) operates five community clinics across Hong Kong (Chai Wan, Tseung Kwan O, Sha Tin, Tuen Mun, and Tsing Yi). These clinics offer outpatient services such as general consultations, vaccinations, and basic health screenings. Currently, each clinic manages appointments and walk-in queues with separate spreadsheets and manual processes, resulting in inefficiencies, long wait times, and limited visibility of service utilisation.

To resolve these issues, CCHC plans to implement a centralised, web-based system called the CCHC Community Clinic Appointment & Queue System. This system will unify appointment management across all clinics, enhance queue transparency, minimise missed appointments, and offer operational reports.

Your development team has been tasked with creating a prototype enterprise web application using Java EE / Jakarta EE technologies. If successful, this prototype will serve as the foundation for a fully operational system.

Driving Question

How can a centralised clinic appointment and queue management system be designed and implemented using enterprise web technologies to improve patient experience, operational efficiency, and service accountability while ensuring secure and controlled access?

System Users and Roles

The implementation of the CCHC Community Clinic Appointment & Queue System aims to support the following users: Patients, Clinic Staff, and Administrators. Public users will be required to register as patients. Clinic Staff and Administrators will be created by the system administrator.

Patients use the system to:

- Public users must register before accessing patient functions.
- Register/log in
- View clinic services and available timeslots
- Book, reschedule, and cancel appointments
- Join a same-day **walk-in queue** (if the clinic accepts walk-ins)
- View personal booking records and queue status
- Receive system notifications (e.g., appointment confirmed, reminder, cancellation outcome, missed appointment)
- Update password and personal profile information

Clinic Staff (Front Desk / Nurse) use the system to:

- View daily appointment lists and walk-in queues by clinic and service
- Approve or reject certain booking requests (e.g., limited quota services)
- Manage check-in (arrived / no-show) and queue progression (call next, skip, completed)
- Record basic visit outcomes (e.g., "Completed", "No-show", "Cancelled by clinic")
- Report operational issues (e.g., doctor unavailable, service suspended)

Administrator (System Administrator / Clinic Manager)

- Create and manage user accounts (Patient / Staff / Admin)
- Configure clinics, services, opening hours, and capacity rules (quota per service per timeslot)
- View analytics and generate reports (per clinic/service/month)
- Review incident logs (e.g., repeated no-shows, frequent cancellations)
- Enforce policy settings (e.g., max active bookings per patient)

Core Function Requirements

A. Appointment Booking and Management

1. Clinic & Service Listing
   - Display list of clinics and services
   - Filter by clinic, service type, date, and availability

2. Book Appointments
   - Patients select a clinic, service, date, and available time slot
   - System prevents double bookings (per patient and per slot capacity)
   - Booking confirmation page and record saved

3. Rescheduling and Cancellation
   - Patients reschedule within allowed rules (e.g., before cutoff time)
   - Patients cancel bookings, freeing the slot
   - Staff can cancel on behalf of the clinic (with reason stored)

4. Check-in and Attendance
   - Staff can mark "Arrived", "Completed", or "No-show"
   - Patients can view appointment status

B. Walk-in Queue Management

1. Join Queue
   - Patients join same-day queue for a clinic/service if enabled
   - System assigns a queue number and estimated waiting time (simple estimate acceptable)

2. Queue Progression

   - Staff call next, skip, or mark as served
   - Patients view live (or near real-time) queue status (refresh-based acceptable)

3. Queue Rules
   - Prevent multiple active queue tickets for the same patient in the same clinic/service/day

C. Notifications

1. Implement at least three types of notifications shown after login :
- Appointment confirmed or updated
- Appointment or clinic cancelled
- Reminder for upcoming appointment
- Queue called, skipped, or expired (any relevant set)

Ensure notifications are role-specific (Patient vs Staff vs Admin dashboards)

D. Analytics / Reporting (for Administrators)

1. View appointment records with filters
- e.g. Filter by clinic, service, month/year, status (completed, no-show, cancelled)

2. Utilisation Rate
- Booking Rate by clinic and service (month/year)
- Example: booked slots ÷ total offered slots for selected clinic/service/month

3. No-show Summary
- Count of no-shows by clinic/service/month (simple table acceptable)
- Account and Role Management

E. Account and Role Management

1. User Management (Admin)
- List users
- Create, edit, or delete users
- Assign roles and clinic links for staff (if applicable)

2. Profile Management (All roles)
- Update passwords and personal details

3. Login Control
- Role-based access to pages/features
- Session management to secure restricted pages

F. Extra Feature

You are expected to work on one of the extra features:
- Batch import services/timeslots/users (CSV upload)
- Basic charts for utilisation/no-shows
- Admin policy settings page (max bookings, cancellation cutoff, queue enable/disable)
- Email-style formatting of notifications
- Audit trail table for staff actions

Description of Tasks

Task	Title	Description
1	Requirements Analysis and System Design	Analyse the scenario and define functional requirements. Design sitemap, navigation flow, and database schema (ERD). Define system rules and constraints.
2	MVC Architecture Implementation	Develop the system using Servlets (Controller), JSP (View), and JavaBeans/DAO (Model). Ensure proper separation of responsibilities.
3	Authentication and Access Control	Implement login, logout, session management, and role-based access control. Protect restricted resources.
4	Appointment Management	Implement booking, rescheduling, cancellation, and attendance management features.
5	Queue Management	Implement queue joining, queue tracking, and queue management functionality.
6	Notification System	Implement database-driven notifications and dashboard display.
7	Administration and Reporting	Implement user management, system configuration, and reporting features.

Project Requirement

Based on the scenario above, you are required to design and develop a web application using Jakarta EE 10 features to address the specified needs. You must form a project group with two members, and each student will specify their own part of the individual work.

Students should share the workload evenly. The group should list down work done by each student.

Work break down	Student 1	Student 2	AI's Contribution

The project will be marked according to the following criteria.

Skills requirements

1. Use JSP/servlets to dynamically generate HTML pages
2. Use JSP/servlets to accept user inputs from browser
3. Use JSP Action
4. Use Custom Tag (Taglib).
5. Use JavaBean
6. Use JDBC for database connection
7. Use session checking
8. Use login control
9. Apply the MVC model

Functionalities and Web design

- Complete the user requirements
- Consistent design and easy to use
- Smooth navigation with the application
- Tidy Page Layout with logical and related graphics
- Error-free implementation
- Creativity

Assessment Criteria

Students will be assessed based on:
1. Technical Implementation
Correct use of required technologies Proper MVC implementation Database integration
2. Functional Completeness and Design
Fulfilment of system requirements Usability and navigation Interface design and consistency
3. Written Report
The report must include: Requirements analysis Sitemap MVC architecture explanation Database design System design description Project schedule Conclusion Skill checklist
4. Demonstration Requirement
Students must demonstrate their system during the lab session. You may be required to: Recompile and run your system Explain your MVC implementation Explain database design and logic Answer technical questions Failure to demonstrate will result in failure of the assignment.

Guideline

Plagiarism

The submitted assignment must be genuinely produced and completed solely by the group members. Plagiarism will be taken very seriously. Any assignments found to be involved wholly or partly in plagiarism, whether from the original authors or from plagiarists, will receive a zero mark.

AI Policy

You are required to read and follow the policy for the use of AI from the student handbook. Generally, it is considered plagiarism if you directly copy any content generated by AI.

Submission of Assignment Work

1. The front page of your submission should include the course title, module title, student identity
2. number, student name, and group number.

3. A written report should include the following:
   a) Assumption and the user and system requirements
   b) Site map
   c) System structure on how the MVC Model is applied
   d) Database structure
   e) Brief description (1 or 2 pages only) on the major characteristics and design of your application
   f) Project Schedule
   g) Conclusions
   h) Skill checklist, which lists your used skills (or technologies) on a single page and highlights the skills and technologies applied in your project

4. Upload all related documents and software project to moodle.vtc.edu.hk on or before the deadline.

5. You are required to demonstrate your assignment. You will receive zero marks for this assignment if you do not demonstrate the assignment in the lab session as required.

- The End -